Author and Year	Paper Title	Dataset and Image Type	Method	Reported Results	Limitation / Gap	URL
Islam et al. (2025)	Rice Variety Classification Using Next Generation Convolutional Networks	Two rice-image datasets	CNN with Transformer-inspired design	99.60% and 100% accuracy on two datasets	Performance on the target PRBD processed-rice dataset is not established	https://doi.org/10.1049/tje2.70102
Farahnakian et al. (2024)	A Comparative Study of State-of-the-art Deep Learning Architectures for Rice Grain Classification	75,000 images, five rice categories	ResNet, VGG, EfficientNet, MobileNet	EfficientNet achieved 99.67% accuracy	Results may not transfer directly to microscopic processed-rice images	https://doi.org/10.1016/j.jafr.2023.100890
Shuvo et al. (2026)	Image-based Classification of Bangladeshi Rice Varieties Using Deep CNNs	15,000 images, five Bangladeshi varieties	VGG16, VGG19, ResNet50, Xception, DenseNet, custom CNN	VGG19: 99.73%; custom model: 98.60%	The target PRBD processed-rice dataset needs separate evaluation	https://doi.org/10.1007/s44462-026-00061-9
Tahsin et al. (2024)	An Extensive Image Dataset for Deep Learning-based Classification of Rice Kernel Varieties in Bangladesh	38 varieties; 19,000 original and 57,000 augmented microscopic images	Dataset creation and image augmentation	Dataset contribution; no model-comparison accuracy benchmark	A model benchmark is still needed for the target classification task	https://doi.org/10.1016/j.dib.2024.111109
Arnab et al. (2025)	RiceKernelEngine: Benchmarking Transfer Learning Models for Microscopic Images of Rice Kernel	3,155 microscopic images, five varieties	VGG19, MobileNetV3, EfficientNet, ConvNeXt	EfficientNet: 96%; MobileNetV3: 94%; ConvNeXt: 87%; VGG19: 85%	Performance on the PRBD dataset remains to be tested	https://doi.org/10.1016/j.array.2025.100429